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Revision 2021 • Semester 1 & 2

Engineering Workshop Practice

Course Code: 2009 Credits: 1.5 Category: Engineering Science Total Hours: 90

Course Overview

Engineering Workshop Practice is a foundational course designed to provide hands-on experience in various engineering trades. The course bridges the gap between theoretical concepts and practical fabrication.

Teaching & Assessment Scheme

Component	Details
Periods per week	3 Practical (P)
Total Periods	45 (Sem 1) + 45 (Sem 2) = 90 Hours
Assessment	Continuous Internal Assessment (CIA) & End Semester Exam (ESE)

Course Objectives:

- Familiarize safety precautions in the workplace.
- Prepare drawings and models for fabrication.
- Identify measuring, marking, holding, striking, and cutting tools.
- Practice electrical wiring, soldering, and motor connections.
- Operate machines and power tools safely.

Course Outcomes (CO)

CO No.	Description	Cognitive Level
CO1	Identify safety precautions, tools and devices required to make carpentry joints.	Applying
CO2	Make use of various tools, machines, instruments and power tools used in the Fitting shop to make fitting joints.	Applying
CO3	Make use of various tools, machines, instruments and power tools used in the Welding shop to make welding joint.	Applying
CO4	Utilize different sheet metal tools and measuring instruments to make sheet metal joints.	Applying
CO5	Make use of various tools and accessories to practice electrical wiring, motor connection and soldering.	Applying

Module 1: Carpentry Shop

18 Hours

Carpentry involves the process of cutting, shaping, and installing building materials during the construction of buildings, ships, timber bridges, etc.

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1.1 Safety Precautions

- **Personal Safety:** Wear tight-fitting clothes, use safety shoes, and avoid loose hair or jewelry.
- **Machine Safety:** Do not operate power tools without permission. Ensure guards are in place.
- **Job Safety:** Keep the workbench clean. Use sharp tools only.

1.2 Carpentry Tools

Category	Tools
Marking & Measuring	Steel rule, Try square, Bevel square, Marking gauge, Mortise gauge, Wing compass.
Cutting Tools	Hand saw, Tenon saw, Firmer chisel, Mortise chisel, Jack plane (Wooden/Metal).
Striking Tools	Mallet, Claw hammer.
Holding Devices	Carpenters Bench vice, G-clamp, Sash clamp.

1.3 Practice Exercise: Open Halved Joint

The halved joint is a woodworking joint in which the two members are joined by removing half the thickness of each at the point of intersection.

Basic representation of a Halved Joint preparation.

Module 2: Fitting Shop

18 Hours

Fitting is the process of assembling various parts manufactured in a machine shop to form a final machine or mechanism.

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2.1 Fitting Tools

- **Marking:** Scriber, Centre punch, Prick punch, Surface plate, V-block, Angle plate.

- **Cutting:** Hack saw (Solid/Adjustable), Chisels (Flat, Cross-cut).
- **Filing:** Single cut and Double cut files. Types: Flat, Square, Round, Half-round, Triangular.
- **Measuring:** Vernier height gauge, Outside/Inside calipers, Jenny caliper.

2.2 Practice Exercise: V-Joint / L-Joint

Students practice filing two metal pieces to achieve a perfect fit in 'V' or 'L' shapes. Precision is checked using a Try Square.

> **Safety Note:** Never use a file without a handle. Do not blow metal chips with your mouth; use a brush.

Module 3: Welding Shop

18 Hours

Welding is a fabrication process that joins materials, usually metals, by using high heat to melt the parts together and allowing them to cool, causing fusion.

3.1 Arc Welding Equipment

- **Welding Machine:** AC Transformer or DC Rectifier.
- **Electrode Holder:** To hold the welding rod.
- **Earth Clamp:** To complete the electrical circuit.
- **Protective Gear:** Face shield/Helmet (with dark glass), Leather Apron, Gloves.

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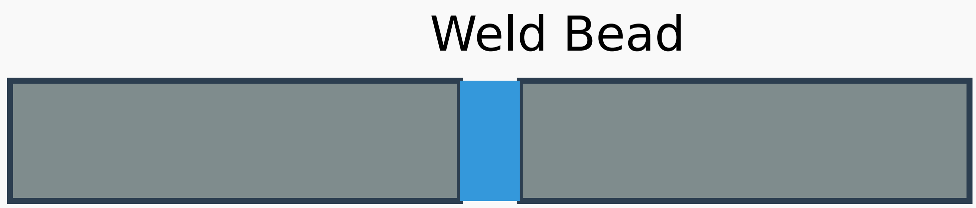
3.2 Welding Practice

Butt Joint: The two workpieces are placed in the same plane and joined at their edges.



Square Butt Joint configuration.

Module 4: Sheet Metal Shop

18 Hours

Sheet metal work involves working with thin metal sheets (usually less than 6mm) to create objects like trays, funnels, and ducts.

4.1 Tools & Operations

- **Snips:** Straight snips and Bent snips for cutting.
- **Stakes:** Metal anvils used for bending and forming (e.g., Hatchet stake, Blow-horn stake).
- **Measuring:** Standard Wire Gauge (SWG) to measure sheet thickness.

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Standard Wire Gauge (SWG): Note that as the Gauge Number increases, the actual thickness of the sheet decreases. Example: 18 SWG is thicker than 24 SWG.

Module 5: Electrical & Soldering

18 Hours

This module covers basic domestic wiring and industrial motor connections.

5.1 Wiring Circuits

- **One Lamp controlled by One Switch:** Basic series connection of a switch and a bulb.
- **Staircase Wiring:** One lamp controlled by two 2-way switches from different locations.
- **DOL Starter:** Direct-On-Line starter for starting induction motors safely.

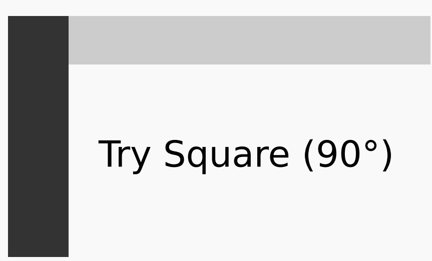
5.2 Soldering

Joining electronic components using a filler metal (solder) and a heat source (soldering iron).

Formula Bank

Concept	Formula	Units
Ohm's Law	V = I × R	V (Volts), I (Amps), R (Ohms)
Electrical Power	P = V × I	P (Watts)
Energy Consumption	E = P × t	E (kWh)

Visual Library



Try Square for checking perpendicularity.

Standard Flat File.

Module-wise Practice Questions

▶ **Q1: What is the use of a Marking Gauge in Carpentry? (3 Marks)**

▶ **Q2: Differentiate between Prick Punch and Centre Punch. (3 Marks)**

▶ **Q3: List the safety equipment used in Welding. (5 Marks)**

Practice Model Paper

Practice Model Paper — Not an Official Examination Paper

Course: Engineering Workshop Practice (2009)

Time: 3 Hours | **Max Marks:** 75 (Internal/External split varies)

Part A: Short Answer Questions (Answer all)

1. Define "Halving" in carpentry.
2. What is the purpose of a V-block in fitting?
3. Name two types of welding joints.
4. What is the function of a Snip in sheet metal work?
5. Expand DOL in motor starters.

Part B: Practical Tasks

1. Prepare an Open Halved Joint from the given wooden piece of size 150x50x25mm.
2. Fabricate a V-joint in the given MS flat piece to the specified dimensions.
3. Perform a straight line bead welding on the given MS plate.

Quick Revision

- **Carpentry:** Focus on grain direction and sharp chisels.
- **Fitting:** Precision is key; use the Try Square frequently.
- **Welding:** Maintain a constant arc length (approx. 3mm).
- **Sheet Metal:** Use a mallet for bending to avoid surface damage.
- **Electrical:** Always check for continuity before powering up.

Glossary

Burr
: A rough edge or ridge left on metal after cutting or drilling.

Flux
: A chemical cleaning agent used in soldering and welding to prevent oxidation.

Kerf
: The width of the cut made by a saw blade.

Stake
: A small anvil used in sheet metal forming.

Official Source Declaration

This lesson content is architected based on the **Revision 2021 Syllabus** for Course Code **2009 (Engineering Workshop Practice)** issued by the State Institute of Technical Teachers' Training & Research (SITTTR), Kerala.

Reference Texts:

- S.K. Hajara Chaudhary, Workshop Technology, Median Promoters.
- J.B. Gupta, Electrical Installation Estimation and Costing, Kataria & Sons.

[illegible] / Malayalam learning support

> [illegible] (Measuring), [illegible] (Marking) [illegible].